$\mathbb{Q}$ 的序结构

定义（$\mathbb{Q}$的序结构）对$\forall [r,s] \in \mathbb{Q}$（其中$r \in \mathbb{Z}$，$s \in \mathbb{Z} \setminus \{0\}$）定义：

$$[r,s] \geqslant 0 \iff rs \geqslant 0$$

对$\forall x, y \in \mathbb{Q}$，定义：

$$x \geqslant y \iff x - y \geqslant 0$$

定义（$\mathbb{Q}$上的绝对值运算）对$\forall x \in \mathbb{Q}$，定义：

$$|x| = \begin{cases} x, & \text{当} x \geqslant 0 \text{时} \\ -x & \text{当} x < 0 \text{时} \end{cases}$$

Lemma：对$\forall x = [r,s] \in \mathbb{Q}$，有：$x = 0 \iff r = 0$

Proof：($\Leftarrow$) $\because r = 0$ $\quad \therefore x = [r,s] = [0,s] = 0$

($\Rightarrow$)：任取$\alpha \in \mathbb{Z} \setminus \{0\}$，则有：$0 = [0, \alpha]$

$\because x = 0$ $\quad \therefore [r,s] = [0,\alpha]$ $\quad \therefore (r,s) \sim (0,\alpha)$

$\therefore r\alpha = 0s = 0$ $\quad \therefore r = 0$ 或 $\alpha = 0$

$\because \alpha \in \mathbb{Z} \setminus \{0\}$ $\quad \therefore r = 0$ $\quad \square$

Lemma：对$\forall x = [r,s] \in \mathbb{Q}$，有：$x \geqslant 0 \iff rs \geqslant 0$

Proof：由定义立得. $\square$

Lemma：对$\forall x = [r,s] \in \mathbb{Q}$，有：$x > 0 \iff rs > 0$

Proof：$x > 0 \iff x \geqslant 0$ 且 $x \neq 0$

$\Longleftrightarrow rs \geqslant 0$ 且 $r \neq 0$

$\overset{①}{\Longleftrightarrow} rs > 0$

①($\Rightarrow$): $\because r \neq 0 \quad \therefore r \in \mathbb{Z}\setminus\{0\} \quad \because s \in \mathbb{Z}\setminus\{0\} \quad \therefore rs \in \mathbb{Z}\setminus\{0\}$

$\because rs \geqslant 0$ 且 $rs \neq 0 \quad \therefore rs > 0$

①($\Leftarrow$): $\because rs > 0 \quad \therefore rs \geqslant 0$.

假设 $r=0$，则有 $rs = 0\cdot s = 0 \quad \therefore 0 > 0$ 矛盾. $\therefore r \neq 0$

$\therefore rs \geqslant 0$ 且 $r \neq 0 \quad \square$

Lemma: 对 $\forall x \in \mathbb{Q}$，有：$x \leqslant 0 \Longleftrightarrow -x \geqslant 0$

Proof: $x \leqslant 0 \Longleftrightarrow 0 \geqslant x$

$\Longleftrightarrow 0 - x \geqslant 0$

$\Longleftrightarrow 0 + (-x) \geqslant 0$

$\Longleftrightarrow -x \geqslant 0 \quad \square$

Lemma: 对 $\forall x = [r,s] \in \mathbb{Q}$，有：$-x = [-r,s] = [r,-s]$

Proof: $\because x + [-r,s] = [r,s] + [-r,s] = [rs + (-r)s, s^2] = [0, s^2] = 0$

$\therefore -x = [-r,s]$

$\because (-r)(-s) = rs \quad \therefore (-r,s) \sim (r,-s)$

$\therefore [-r,s] = [r,-s]$

$\therefore -x = [-r,s] = [r,-s] \quad \square$

Lemma: 对$\forall x=[r,s]\in\mathbb{Q}$，有: $x\leqslant 0 \iff rs\leqslant 0$

Proof: $x\leqslant 0 \iff -x\geqslant 0$

$\iff [-r,s]\geqslant 0$

$\iff (-r)s\geqslant 0$

$\iff -(rs)\geqslant 0$

$\iff rs\leqslant 0$ $\square$

Lemma: 对$\forall x=[r,s]\in\mathbb{Q}$，有: $x=0 \iff rs=0$

Proof: $(\Rightarrow)$: $\because x=0$ $\therefore r=0$ $\therefore rs=0\cdot s=0$

$(\Leftarrow)$: $\because r\in\mathbb{Z}, s\in\mathbb{Z}\setminus\{0\}$，$rs=0$ $\therefore r=0$或$s=0$

$\because s\neq 0$ $\therefore r=0$ $\therefore x=0$ $\square$

Lemma: 对$\forall x=[r,s]\in\mathbb{Q}$，有: $x<0 \iff rs<0$

Proof: $x<0 \iff x\leqslant 0$ 且 $x\neq 0$

$\iff rs\leqslant 0$ 且 $rs\neq 0$

$\iff rs<0$ $\square$

Lemma: 对 $\forall x \in \mathbb{Q}$, ① $x=0$ ② $x<0$ ③ $x>0$ 有且仅有一个成立.

Proof: $\because x \in \mathbb{Q}$ $\therefore \exists r \in \mathbb{Z}, s \in \mathbb{Z} \setminus \{0\}$, s.t. $x=[r,s]$

$\because r \in \mathbb{Z}$ 且 $s \in \mathbb{Z}$ $\therefore rs \in \mathbb{Z}$

$\therefore$ ① $rs=0$ ② $rs<0$ ③ $rs>0$ 有且仅有一个成立

$\Updownarrow$ $\Updownarrow$ $\Updownarrow$

$\therefore$ ① $x=0$ ② $x<0$ ③ $x>0$ 有且仅有一个成立. $\square$

Lemma: 对 $\forall x, y \in \mathbb{Q}$, 有: $x \geqslant y \iff x-y \geqslant 0$

Proof: 由定义立得. $\square$

Lemma: 对 $\forall x, y \in \mathbb{Q}$, 有: $x=y \iff x-y=0$

Proof: $(\Longrightarrow)$: $x-y=x+(-y)=y+(-y)=0$

$(\Longleftarrow)$: $y=0+y=(x-y)+y=(x+(-y))+y=x+((-y)+y)$

$=x+0=x$

$\therefore x=y$ $\square$

Lemma: 对 $\forall x, y \in \mathbb{Q}$, 有: $x>y \iff x-y>0$

Proof: $x>y \iff x \geqslant y$ 且 $x \neq y$

$\iff x-y \geqslant 0$ 且 $x-y \neq 0$

$\iff x-y>0$ $\square$

Lemma: 对 $\forall x \in \mathbb{Q}$，有：$x=0 \iff -x=0$

Proof: $(\Rightarrow)$: $\because 0+0=0$ $\quad \therefore -0=0$

$\therefore -x=-0=0$

$(\Leftarrow)$: $x=-(-x)=-0=0$ $\quad \square$

Lemma: 对 $\forall x \in \mathbb{Q}$，有：$x<0 \iff -x>0$

Proof: $x<0 \iff x \leq 0$ 且 $x \neq 0$

$\iff -x \geq 0$ 且 $-x \neq 0$

$\iff -x>0$ $\quad \square$

Lemma: 对 $\forall x \in \mathbb{Q}$，有：$x>0 \iff -x<0$

Proof: $\because x \in \mathbb{Q}$ $\quad \therefore -x \in \mathbb{Q}$

$\therefore -x<0 \iff -(-x)>0 \iff x>0$ $\quad \square$

Lemma: 对 $\forall x, y \in \mathbb{Q}$，有：$x<y \iff x-y<0$

Proof: $x<y \iff y>x$

$\iff y-x>0$

$\iff -(y-x)<0$

$\iff x-y<0$ $\quad \square$

Lemma：对 $\forall x, y \in \mathbb{Q}$，① $x=y$ ② $x<y$ ③ $x>y$ 有且仅有一个成立.

Proof：$\because x, y \in \mathbb{Q}$ $\therefore x-y \in \mathbb{Q}$

$\therefore$ ① $x-y=0$ ② $x-y<0$ ③ $x-y>0$ 有且仅有一个成立

$\Updownarrow$ $\Updownarrow$ $\Updownarrow$

$\therefore$ ① $x=y$ ② $x<y$ ③ $x>y$ 有且仅有一个成立. $\square$